

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
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17. Perform Edge detection using Sobel Matrix along X axis.

AIM:

Edge detection using Sobel Matrix along X axis.

PROGRAM:

```
import cv2

img = cv2.imread(r'C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png', 0)

if img is None:
    print("Error: Image not found!")
else:
    # Apply Sobel filter in the X direction
    sobel_x = cv2.Sobel(img, cv2.CV_8U, 1, 0, ksize=5)

    # Save the output image
    cv2.imwrite(
        r'C:\Users\Susmitha\Downloads\ITA0510 Lab\sobel_x.jpg',
        sobel_x
    )

    # Display the output
    cv2.imshow("Original Image", img)
    cv2.imshow("Sobel X", sobel_x)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.